

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a9a3254a595af2a6d.jsonl`  
- SHA-256: `2aaf3a0f43f7741e89dd9864c52f9a294c6f999d6b965f23776a84c670b2b858`  
- Source modified: `2026-06-06T17:34:13+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:32:36.252Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Badminton point/rally segmentation can be done by a per-frame CNN image classifier that distinguishes broadcast-camera rally frames from non-rally frames (replays/close-ups), achieving F1 97.23 / Precision 98.06 / Recall 92.34 ? without any shuttle tracking."

****Source:**** <https://arxiv.org/pdf/2308.12199> (primary)

****Supporting quote:**** "Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups. ... CNN: F1-Score 97.23, Precision 98.06, Recall 92.34"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:32:40.794Z)

I'll verify this claim about per-frame CNN image classification for badminton rally segmentation. Let me start by examining the primary source.

3. assistant (2026-06-06T17:32:44.318Z)

Let me fetch the primary source to verify the exact claim and numbers.

4. user (2026-06-06T17:32:48.446Z)

maxContentLength size of 10485760 exceeded

5. user (2026-06-06T17:32:53.722Z)

Paper Summary

****Title:**** "Towards Real-Time Analysis of Broadcast Badminton Videos"

****Authors:**** Nitin Nilesh, Tushar Sharma, Anurag Ghosh, C. V. Jawahar

****Publication:**** arXiv:2308.12199 (Computer Vision and Pattern Recognition), submitted August 23, 2023

****Topic:**** Computer Vision and Pattern Recognition (cs.CV)

Abstract Summary

The paper presents an end-to-end framework analyzing player movements in badminton matches using only visual data from live broadcasts. The system "remove[s] replays and other redundant parts

of the broadcast match" through gameplay segmentation, then performs player tracking to measure on-court distance and movement speed. It generates heatmaps showing player coverage areas. The framework was successfully deployed during Premier Badminton League 2019.

Key Elements

****Yes, it involves:****

- ****Badminton analysis**** - primary focus
- ****Rally segmentation**** - removes non-gameplay content
- ****Player tracking**** - identifies and tracks both players

****No explicit mention of:****

- CNN frame classification
- Shuttle tracking

The approach relies on visual cues rather than multi-modal sensor data and achieved real-time performance during live broadcasts.

6. assistant (2026-06-06T17:32:56.264Z)

The abstract confirms the paper topic. Now let me get the specific details about the CNN classifier and the exact metrics. Let me try the HTML version of the paper.

7. user (2026-06-06T17:33:01.682Z)

I cannot find the section on gameplay segmentation, the rally vs non-rally classification method, or the specific performance metrics you're asking about (F1 97.23, Precision 98.06, Recall 92.34) in the provided content.

The webpage excerpt contains only the abstract and metadata for the arXiv paper. The abstract mentions that the authors "focus on the gameplay by removing replays and other redundant parts of the broadcast match," but it does not provide details about the classification methodology, model architecture, or performance numbers.

To access this information, you would need to view the full PDF of the paper itself, which is linked on the page but not included in the content you provided.

8. user (2026-06-06T17:33:05.222Z)

Web search results for query: "Towards Real-Time Analysis of Broadcast Badminton Videos" gameplay segmentation CNN rally non-rally F1 precision recall"

Links: [{"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.researchgate.net/publication/322075902_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos | Request PDF", "url": "https://www.researchgate.net/publication/325002967_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos | DeepAI", "url": "https://deepai.org/publication/towards-real-time-analysis-of-broadcast-badminton-videos"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://cvit.iiit.ac.in/research/projects/cvit-projects/badminton-analytics"}, {"title": "(PDF) Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://www.academia.edu/70243179/Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}]

Based on the search results, I found information about "Towards Real-Time Analysis of Broadcast Badminton Videos." Here's what I found:

Overview

This work proposes an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos. The approach uses only visual cues rather than multi-modal sensor

data.

Key Components

The framework focuses on gameplay by removing replays and redundant parts, then performs player tracking to identify and track player movements, finally calculating the distance covered by each player and their average speed on the court.

Performance and Application

The proposed framework was successfully used to analyze live broadcast matches in real-time during the Premier Badminton League 2019 (PBL 2019), with commentators and broadcasters appreciating the utility.

Related Work

There is also related research on structured badminton video analysis that addresses gameplay segmentation. One approach segments the points played, tracks and recognizes players, and annotates their strokes, achieving 95.44% point segmentation accuracy, 97.38% player detection score (mAP@0.5), and 97.98% player identification accuracy.

The papers use CNN-based approaches and temporal segmentation techniques for analyzing badminton matches, though the specific metrics you mentioned (F1 precision, recall for rally vs. non-rally classification) are not detailed in these search results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:33:10.872Z)

There's an important nuance emerging: there are two related papers from the same group (CVIT/IIIT-Hyderabad). Let me get the actual PDF content. The PDF is too large for WebFetch, so let me download it and read the relevant section directly.

10. user (2026-06-06T17:33:13.392Z)

```
-rw-r--r-- 1 avidu 197609 11475365 Jun  6 23:03 paper.pdf
paper.pdf: PDF document, version 1.5
```

11. assistant (2026-06-06T17:33:16.086Z)

The PDF is 11MB. Let me read it directly to find the segmentation section and exact metrics.

12. user (2026-06-06T17:33:16.109Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

13. assistant (2026-06-06T17:33:19.589Z)

The Bash tool runs in a Linux sandbox (WSL) but Read uses Windows paths. Let me extract the text from the PDF in the Bash environment.

14. user (2026-06-06T17:33:21.814Z)

pdftotext OK

15. user (2026-06-06T17:33:25.342Z)

16:Abstract--Analysis of player movements is a crucial subset of sports analysis. Existing player movement analysis methods use recorded videos after the match is over. In this work, we propose an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos. We only use the visual inputs from the match and, unlike other approaches which use multi-modal sensor data, our approach uses only visual cues. We propose a

method to calculate the on-court distance covered by both the players from the video feed of a live broadcast badminton match. To perform this analysis, we focus on the gameplay by removing replays and other redundant parts of the broadcast match. We then perform player tracking to identify and track the movements of both players in each frame. Finally, we calculate the distance covered by each player and the average speed with which they move on the court. We further show a heatmap of the areas covered by the player on the court which is useful for analyzing the gameplay of the player.

17:Our proposed framework was successfully used to analyze live broadcast matches in real-time during the Premier Badminton League 2019 (PBL 2019), with commentators and broadcasters appreciating the utility.

22:Fig. 1: An overview of the distance travelled by the player from the start of a set. Player in blue bounding box refers to the "top-player" and player in yellow bounding box refers to the "bottom-player". Dot at the center-bottom of the player's bounding box represents their current location. The "Rally Duration" is the time length of a particular rally. Four such rallies are depicted for reference. (best viewed in color)

26:

2. Many of the existing analysis methods are constrained by these challenges. Our main focus in this work is to provide real-time analysis framework for the broadcast videos which is expected to generalize to almost all the badminton games captured from broadcast camera view.

27:In this work, we introduce an approach which generates heatmaps of player movements, calculates distance run by players and the average speed of player movements for the game of badminton (figure 1). We perform point segmentation, player detection and homography based techniques to compute the above defined metrics.

29:1) We propose an end-to-end framework (figure 3) to analyze live broadcast match videos, the components of which can be trained using pre-recorded badminton matches. Unlike previous approaches, our method uses visual cues only and does not rely on any hardware setup.

31:3) We tested our framework in a real-time scenario and analyzed live broadcast matches that were conducted as a part of the Premier Badminton League 2019.

32:4) We introduce two new datasets, one each for singles and doubles matches covering a wide range of broadcasting camera angles, court and lighting conditions, and featuring several different players. We also introduce a set of 25 rallies badminton match video of our own recording having the ground truth data for the distance covered by the players in each rally.

37:only visual cues and does not rely on special camera setup or additional sensors. There has also been some work in analysing sports without reliance on anything other than the broadcast video. Sukhwani et al. [19] and Ghosh et al. [20] computed frame level annotations in tennis videos, however, where Sukhwani et al. used a dictionary learning method to co-cluster available textual descriptions, Ghosh et al. [20] used the scoreboard extraction approach to perform the index based analysis.

38:Real-time Sports Analysis: Zhong et. al. [21], [22] present a real-time framework for scene detection and structure analysis for tennis and baseball games using compressed-domain processing techniques. Further, they introduce another realtime framework to detect the syntactic structures that are at a level higher than shots for the game of tennis. Ekin et al. [23] propose a single generic real-time algorithm to detect playbreak events for multiple sports (football, tennis, basketball, and soccer). Their algorithm uses shot-based generic cinematic features to detect the events.

39:Our work is inspired by the work of [4] that proposes an end-to-end framework to automatically annotate badminton broadcast videos. Their work identifies various understandable metrics that can be computed for analyzing badminton matches which in turn help qualitative understanding of badminton games. We build on top of [4], and extend the work further by (a) covering wider match situations/tournaments/games (b) adapting the solution to real-time deployment and (c) a successful field trial in PBL 2019 with commentators and broadcasters appreciating the utility.

48:

Fig. 2: Our dataset contains various broadcast camera positions, illuminations settings, different backgrounds and court colors which helps generalize the framework for any kind of badminton match scenario. It should be observed that having audiences in the background may affect the player localization method if not generalized well.

65:To train and validate our approach, we took 10 matches each for both the singles and doubles matches, which contains 22 hours 16 minutes data, and annotate them for point segments and player bounding boxes. We split the ten matches into training set and testing set of 7 and 3 matches, respectively. We plan to release our dataset and annotations publicly post acceptance of the work.

67:A. Point Segmentation Annotation

68:Broadcast badminton matches are divided into gameplays and replays. We define a "rally" frame as a frame where the actual gameplay is happening and a "non-rally" frame as a frame where the

replay (highlights) has been shown, or players are resting, etc. (or in other words whether the gameplay is not happening). Generally, non-rally frames contain redundant information about the game and do not contribute much in the analysis part. Therefore, we first start classifying between rally and non-rally frames. We call this classification method as Point Segmentation. This allows us to keep all the rally frames, group them into rallies according to the points scored by the players and discard the non-rally frames. To make the dataset usable for point segmentation, we annotate every frame

70:of a match as a rally or non-rally frame using the ELAN video annotation tool 4. We annotate 23156 frames in total, where 15147 were non-rally frames and 8009 were rally frames.

72:We annotate the dataset to get the bounding box for each player using the LabelImg5 graphical image annotation tool. We define the player's position in the court with respect to the broadcast camera point of view where one player plays from the near side, and other player plays from the far side of the camera, and we define them as bottom and top player respectively (see figure 1 for more details). We manually annotate players' bounding boxes with two classes "PlayerTop" and "PlayerBottom" where "PlayerTop" corresponds to the player on the far side of the court and while "PlayerBottom" corresponds to the player on the near side of the court with respect to the viewpoint of the camera. We randomly pick 150 rally frames from each match and annotated them for player bounding boxes for both the players. Thus, we end up with 3000 (150 frames * 2 players * 10 matches) annotated frames in total. We use this annotation to learn player bounding boxes for the rest of the frames. We use the same annotation style for the doubles matches as well. As shown in figure 4, occasionally, players are occluded or blurred due to the fast-paced nature of the game. Such occlusions are even more prevalent in the doubles matches where there are four players on court as opposed to the two players in a singles match.

77:
Fig. 3: We propose a real-time end-to-end pipeline to analyze live broadcast match videos. Our pipeline consists of point segmentation, player localization, pictorial point summary and top-view space conversion to compute the distance and overall analysis.

79:which were accurate up to 1 meter, measured the distance covered by each player during each rally. We recorded 25 rallies and obtained the distance values for each player for all of them. All the videos were recorded at approximately the same broadcast angle as that of the singles and doubles datasets using a smartphone camera.

81:As shown in figure 3, our pipeline is largely based on the approach proposed by Ghosh et al [4]. It comprises of 4 steps: (1) Point Segmentation (2) Player Localization (3) Top View Space Conversion and (4) Analysis. We optimize each of these steps independently to make them more robust while simultaneously reducing latency. The improved pipeline can be used for real-time analysis of videos.

82:A. Point Segmentation

83:The very first step of our pipeline is to keep the rally frames and discard the non-rally frames (which are redundant for our analysis) from the given match video. Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups of players, additional graphics (score cards) and unusual camera angles. These differences help distinguish these frames. Unlike Ghosh et al [4] who use HOG + SVM, we use a convolutional neural network based classifier to classify rally and non-rally frames. We finetune the ResNet-18 [24] model (pre-trained on ImageNet [25] dataset) by changing the last layer of this network to label the frame as rally frame or non-rally frame. As shown in table II, our CNN based method is both more accurate and also exhibits lower latency compared to HOG + SVM.

84:At test time, we classify the frames retrieved from the match video and number the rally frames in a sequential manner.

86:We assume that a rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies. This allows us to group frames of the match into individual rallies by separating sets of continuous rally frames with at least 3 seconds of non-rally frames in between them. Thus Point Segmentation gives us a set of rallies from the match. We number each rally sequentially as rally-1, rally-2 etc.

88:Once we obtain individual rallies, we need to find player location in each frame of each rally. A YOLOv3 [26] network is finetuned for these two classes. In order to overcome the constraint of only having 3000 annotated samples for player localization, we use transfer learning to train the YOLOv3 network. We first pre-train the YOLOv3 network on PASCALVOC dataset [27] which has 20 classes including the "Person" class. Further, we finetune the YOLOv3 network on our own dataset to achieve better accuracy. We used the same method for both the single's and double's matches to get the location of the player. However, as can be seen in figure 4, player localization is a very difficult problem to solve in doubles matches.

89:Using a convolutional network for point segmentation and YOLOv3 for player localization helps us build a fast and robust pipeline that performs better than [4] both in terms of accuracy of predictions and latency. The improved accuracy as well as inference time make it possible to use

our pipeline for a real-time implementation. Please refer to table II for more details.

96:HOG+SVM CNN

99:95.44 97.23

102:97.83 98.06

105:91.02 92.34

110:Fig. 5: Conversion of player locations from camera coordinates to top view co-ordinates. The top-view space provides the player's position and footwork information around the court for a particular gameplay. This view can also be seen as heatmap of the players for that particular rally. (Best viewed in color)

112:(a) Point Segmentation Method

117:Faster R-CNN YOLOV3

125:Rally Latency (Seconds)

133:side of the camera. In order to find out the distance or other kind of analysis, we need to switch the view of player positions from broadcast camera view to another view where the displacement of the player is not scaled by the near side or far side of the player position. We choose the top view of the badminton court where we map the player locations from broadcast camera view. So, using homography techniques, we map the 2-dimensional broadcast view camera coordinates to top view court coordinates. By doing this, we get an equivalent linear top view for the player locations. To perform homography calculation, we find the court lines from the video frame using Hough line finding method [28] and get the position of court corners. We compute a homography matrix between these four court corners and top view four court corners to get a mapping from camera coordinates to top view space coordinates. Finally, we map the player locations from broadcast camera view to top view. Please refer to figure 5 for output details.

135:Player Distance Calculation: To compute the distance covered by the player in a particular rally, we utilize our player tracks in top view space. Now the player locations are in 2dimensional space and we take the "center bottom" point of the player bounding box as a proxy for the player's current location.

136:According to the standard badminton rules⁶, the length and width of the badminton court is 13.41 meters and 6.1 meters respectively. From this information, we know that from the top left corner of the court to the top right corner of the court, the distance of pixels is 6.1 meters and similarly, from the top left corner of the court to the bottom left corner of the court the distance is 13.41 meters. We use this information to calculate the distance of the player covered in meters. We calculate the euclidean distance between the locations of the player in successive frames for both players. Now we have the pixel-based distance between two successive frames. We use unitary method to scale this pixel-based distance into meters to get the actual distance values. We add all the distances (meters) successively for the whole rally to calculate the distance for each player in that particular rally. The distance calculation of

139:

a rally for one player is formulated as:

165:where n is the number of frames in the rally, x_i and y_i are the co-ordinates of the player in the top-view space at i th frame, D_h is the pixel-based distance from top-left corner to the top-right corner (horizontal distance) and D_v is the pixel-based distance from top-left corner to the bottom-left corner (vertical distance). For the game of double's, we have two players on each side of the court i.e. two "PlayerTop" (pt) and two "PlayerBottom" (pb). Therefore, for the first frame, we initialize pt_1 and pt_2 for the two top players and pb_1 and pb_2 for the two bottom players respectively. In subsequent frames, we use the player's previous position to track them individually. Again, we use the method same as singles to calculate the distance.

167:Player Average Speed Calculation: To calculate the average speed achieved by each of the players in a particular rally, we use the distance covered by the players. As we know the number of frames in a particular rally, we find out the total time elapsed (25 frames per second) and divide the player distance covered by the total time.

171:To evaluate the accuracy of our proposed pipeline in terms of the distance covered by the players in a particular rally, we used our recorded badminton match video which has ground truth data of 25 rallies for the distance covered by the players. We use our proposed pipeline to obtain predicted distance values for each rally and compare them to the ground truth values. We used the same evaluation metric as [29], which uses paired t-tests to calculate differences between the ground truth distances and predicted distances. We report the mean, standard deviation, minimum, and maximum error between the ground truth and predicted values for both "Player Top" and "Player Bottom". We also report the t-stats value and degree of freedom of the paired t-test calculation between the ground truth and predicted values. Please refer to table III for the report.

187:The proposed pipeline was successfully used in a real-world setting to analyze live broadcast matches in real-time during the Premier Badminton League 2019. As reproducing the exact same scenario is not feasible, we use a simulation of the realtime scenario to evaluate the performance (with respect to latency) of our model. Generally, as live matches are streamed,

the camera recording the match captures 25 frames per second and thus the match video file size increases with time. In order to simulate the real-time scenario, we fed the pre-recorded match videos to our proposed pipeline at precisely 25 frames per second, sequentially. Thus our models did not have access to all the frames at the same time. We ran this simulation on 61 rallies (totaling up to 55 minutes) and timed our models. Figure 6 shows the results of this simulation. 189: We use the live feed of the broadcast video match from Premier Badminton League - 2019 (courtesy of Star Sports India⁷). We used frames from the live video in batches of 20 for our experiments. We attempt to extract the analysis from the data we got using the methods mentioned above. We have segmented the frames into either rally or non-rally frame using point segmentation method which in the end, gives us rally directories as soon as the rally is over in the live match, where each directory contains rally frames for that specific rally. Then we pass the rally frames to the player localization method to get the player bounding boxes. We perform homography to change the space from broadcast camera view to top-view. We perform each analysis on frame level for the accessed file.

192:

On an average, one rally is 10 to 12 seconds long and it contains 250 to 300 rally frames. For a real time implementation, it is not feasible to process these many frames in one go. Hence, to solve this problem for this scenario, we process every 3rd rally frame for player localization method. As there are 25 frames per second in the broadcast match video, we assume that there is no significant change in player location if we take every third rally frame. After getting the rally directories from the above point segmentation method we pass the rally directories to the player localization method to get the player location for each of the frames in a parallel manner. Using this rally-level parallelism for analysis helped us to achieve the goal in the real-time scenario.

196: cannot be used to display the analysis or the computed match statistics in real time. In this work, we present an efficient and robust pipeline to analyze badminton matches (both singles and doubles) in real-time. We perform 3 different types of analysis: (1) heatmap generation (2) distance covered by players and (3) average speed of player movement. We optimize each component of our pipeline to ensure that it maintains high accuracy while also minimizing inference latency. We show the evaluations for the player distance calculation and its implementation in real-time scenario. We also empirically show that processing every third frame of the video is sufficient to perform the desired analysis. The real-time implementation of these kinds of analyses not only enhances the viewer experience but also helps players improve their game.

197: Another problem that hinders research in the area of sports analysis, and more specifically, badminton analysis, is the lack of labelled datasets. We introduce 3 different datasets to help combat this problem. Two of our datasets (singles and doubles) consist of Badminton World Federation match videos and feature professional badminton players and official courts and recording equipment. Both these datasets are annotated for Point Segmentation and Player Localization. The third dataset consists of amateur players recorded by using a smartphone camera from the approximately same broadcast angle as the other datasets. This dataset also consists of ground truth data for the distance covered by the players in each rally (which was obtained from the distance trackers worn by the players during gameplay). We hope that access to these datasets helps spur further research into real-time analysis of badminton videos.

223: [19] M. Sukhwani and C. Jawahar, "Frame level annotations for tennis videos," in ICPR. IEEE, 2016. 2

16. assistant (2026-06-06T17:33:30.340Z)

I now have the full picture. The claim is well-supported by the primary source. Let me verify the exact metrics table mapping (lines 96-105) and confirm the precision number, since the claim states Precision 98.06 but I want to verify it's not 97.83.

The table (lines 96-105) shows:

- Line 96: `HOG+SVM | CNN` (column headers)
- Line 99: `95.44 | 97.23` (F1-Score)
- Line 102: `97.83 | 98.06` (Precision)
- Line 105: `91.02 | 92.34` (Recall)

So for the CNN column: F1=97.23, Precision=98.06, Recall=92.34. This matches the claim exactly. Let me confirm the row labels for the table.

17. user (2026-06-06T17:33:31.302Z)

C. Top View Space

We know that the displacement of both the players would manifest differently in the camera coordinates as one player is near side of the camera and another one is on the far

Fig. 4: Output of player tracking method. It should be noted that this method works well, even when there are levels of occlusion and irrespective of player poses.

Method
HOG+SVM CNN

F1-Score
95.44 97.23

Precision
97.83 98.06

Recall
91.02 92.34

Match Latency (Minutes)
57.24 16.75

Fig. 5: Conversion of player locations from camera coordinates to top view co-ordinates. The top-view space provides the player's position and footwork information around the court for a particular gameplay. This view can also be seen as heatmap of the players for that particular rally. (Best viewed in color)

18. assistant (2026-06-06T17:33:38.227Z)

Confirmed verbatim. The table maps exactly. Now let me critically assess the claim's framing against the checklist. The key question for an adversarial review: is the claim an overreach or misread of what the paper actually demonstrates?

Let me check the critical caveats:

1. "without any shuttle tracking" ? TRUE, the method uses ResNet-18 on raw frames, no shuttle.
2. The metrics are **frame-level classification** accuracy (rally vs non-rally frame), NOT rally-segment/boundary detection accuracy at IoU 0.5. This is a meaningful distinction for the research context.
3. The method relies on **broadcast-camera vs close-up/replay distinction** ? works because non-rally frames are visually distinct (close-ups, scorecards, unusual angles).

Let me verify whether the claim conflates frame-level F1 with rally-segment-level F1, which is the most likely refutation angle. Let me check what these numbers actually measure and whether the broadcast-camera assumption is a generalization limitation.

19. user (2026-06-06T17:33:38.956Z)

A. Point Segmentation Annotation

Broadcast badminton matches are divided into gameplays and replays. We define a "rally" frame as a frame where the actual gameplay is happening and a "non-rally" frame as a frame where the replay (highlights) has been shown, or players are resting, etc. (or in other words whether the gameplay is not happening). Generally, non-rally frames contain redundant information about the game and do not contribute much in the analysis part. Therefore, we first start classifying between rally and non-rally frames. We call this classification method as Point Segmentation. This allows us to keep all the rally frames, group them into rallies according to the points scored by the players and discard the non-rally frames. To make the dataset usable for point segmentation, we annotate every frame

of a match as a rally or non-rally frame using the ELAN video annotation tool 4. We annotate 23156 frames in total, where 15147 were non-rally frames and 8009 were rally frames.

B. Player Bounding Boxes Annotation

We annotate the dataset to get the bounding box for each player using the LabelImg5 graphical image annotation tool. We define the player's position in the court with respect to the broadcast camera point of view where one player plays from the near side, and other player plays from the far side of the camera, and we define them as bottom and top player respectively (see

figure 1 for more details). We manually annotate players' bounding boxes with two classes "PlayerTop" and "PlayerBottom" where "PlayerTop" corresponds to the player on the far side of the court and while "PlayerBottom" corresponds to the player on the near side of the court with respect to the viewpoint of the camera. We randomly pick 150 rally frames from each match and annotated them for player bounding boxes for both the players. Thus, we end up with 3000 (150 frames * 2 players * 10 matches) annotated frames in total. We use this annotation to learn player bounding boxes for the rest of the frames. We use the same annotation style for the doubles matches as well. As shown in figure 4, occasionally, players are occluded or blurred due to the fast-paced nature of the game. Such occlusions are even more prevalent in the doubles matches where there are four players on court as opposed to the two players in a singles match.

C. Recorded Badminton Match Video

Our analysis pipeline predicts distance covered by the players, and this forms the basis for other kinds of analysis (like average speed with which the players are moving). Therefore, it is imperative to rigorously evaluate the distance values predicted by our pipeline. Towards this end, we created a new dataset consisting of videos of amateur players playing badminton while wearing distance trackers. The distance trackers,

4<https://tla.mpi.nl/tools/tla-tools/elan/> 5<https://github.com/tzutalin/labelImg>

Fig. 3: We propose a real-time end-to-end pipeline to analyze live broadcast match videos. Our pipeline consists of point segmentation, player localization, pictorial point summary and top-view space conversion to compute the distance and overall analysis.

which were accurate up to 1 meter, measured the distance covered by each player during each rally. We recorded 25 rallies and obtained the distance values for each player for all of them. All the videos were recorded at approximately the same broadcast angle as that of the singles and doubles datasets using a smartphone camera.

IV. METHOD

As shown in figure 3, our pipeline is largely based on the approach proposed by Ghosh et al [4]. It comprises of 4 steps: (1) Point Segmentation (2) Player Localization (3) Top View Space Conversion and (4) Analysis. We optimize each of these steps independently to make them more robust while simultaneously reducing latency. The improved pipeline can be used for real-time analysis of videos.

A. Point Segmentation

The very first step of our pipeline is to keep the rally frames and discard the non-rally frames (which are redundant for our analysis) from the given match video. Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups of players, additional graphics (score cards) and unusual camera angles. These differences help distinguish these frames. Unlike Ghosh et al [4] who use HOG + SVM, we use a convolutional neural network based classifier to classify rally and non-rally frames. We finetune the ResNet-18 [24] model (pre-trained on ImageNet [25] dataset) by changing the last layer of this network to label the frame as rally frame or non-rally frame. As shown in table II, our CNN based method is both more accurate and also exhibits lower latency compared to HOG + SVM. At test time, we classify the frames retrieved from the match video and number the rally frames in a sequential manner.

We assume that a rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies. This allows us to group frames of the match into individual rallies by separating sets of continuous rally frames with at least 3 seconds of non-rally frames in between them. Thus Point Segmentation gives us a set of rallies from the match. We number each rally sequentially as rally-1, rally-2 etc.

B. Player Localization

20. assistant (2026-06-06T17:33:57.478Z)

I now have everything needed for a thorough adversarial assessment. Let me work through the checklist.

****Checklist analysis:****

1. ****Is the claim supported by the quote/source?**** YES, with high fidelity. The paper "Towards Real-Time Analysis of Broadcast Badminton Videos" (arXiv:2308.12199, Nilesh, Sharma, Ghosh, Jawahar, IIIT-Hyderabad CVIT, Aug 2023) describes a ResNet-18 (ImageNet-pretrained, last layer swapped) fine-tuned to classify each frame as rally vs non-rally ("Point Segmentation"). Table II reports CNN: F1=97.23, Precision=98.06, Recall=92.34. All three numbers match exactly. The

supporting quote about broadcast-camera vs close-up is verbatim from Section IV.A. No shuttle tracking is used ? confirmed.

2. **Contradicting evidence?** None found. The numbers are internally consistent and verbatim from the primary source.

3. **Source quality?** Primary source ? peer-group academic paper from a credible CV lab (IIIT-Hyderabad), field-deployed at PBL 2019. Adequate for the claim's strength.

4. **Outdated?** 2023 ? current enough; the architecture (ResNet-18) is mature/standard.

5. **Marketing/cherry-picked?** No.

However ? the adversarial nuances that qualify (but do not refute) the claim:

- The metrics are **frame-level binary classification** accuracy (per-frame rally vs non-rally), NOT rally-segment boundary detection at IoU 0.5. The research context the verifier operates in (WASB pipeline) measures rally-window F1 at IoU 0.5 ? these are different metrics and not directly comparable. The 97.23 F1 is per-frame, not per-rally-segment.
- The task it solves is "discard replays/close-ups from broadcast feed" ? i.e., scene/shot-type filtering. It is NOT detecting the fine-grained start/end of a point within continuous broadcast-camera footage. The actual rally grouping is done by a crude heuristic (?3s rally, ?3s gap), not the CNN.
- It depends entirely on the broadcast-camera-vs-close-up visual distinction. For self-recorded single-fixed-camera footage (the user's local-first context, and the paper's own amateur "recorded" dataset), there are no replays/close-ups to discriminate, so this exact method does not transfer to that within-camera segmentation problem.
- Singles + doubles BWF broadcast data; train/test = 7/3 matches; not released at time of writing.

These are scope qualifications, not falsifications. The claim as literally stated ? "can be done by a per-frame CNN image classifier... achieving F1 97.23 / Precision 98.06 / Recall 92.34 ? without any shuttle tracking" ? is accurate and verbatim-supported. The claim does NOT itself assert IoU-0.5 segment-level performance or transfer to single-camera footage, so it is not an overreach of what's printed.

Per the rubric, refuted=true requires unsupported/contradicted/low-quality/outdated/marketing. None apply. The claim is well-supported, current, primary-sourced. refuted=false, with the important caveat that it measures frame-level (not segment-level/IoU) accuracy on broadcast footage with replays ? a context distinct from the user's single-camera pipeline.

21. user (2026-06-06T17:34:08.004Z)

Structured output provided successfully

22. assistant (2026-06-06T17:34:13.280Z)

Verification complete. The claim is NOT refuted ? all three metrics (F1 97.23 / Precision 98.06 / Recall 92.34) match verbatim against Table II of the primary source (arXiv:2308.12199), and the method description confirms a ResNet-18 frame classifier with no shuttle tracking. I flagged the key scope caveats (frame-level not segment-level/IoU metrics; depends on broadcast-vs-close-up distinction that doesn't exist in single-camera footage) as qualifications, but they don't falsify the literal claim.